

RISK MANAGEMENT FILE

CSI-100 Cell Sorting Instrument (IVD)

Document number	RMF-CSI-001 (includes HA-CSI-001 Rev C, RMR-CSI-001 Rev A)
Revision	C
Document type	Risk management file (ISO 14971:2019)
Status	Released
Effective date	May 2027 (pre-market release)
Author	Atif Lodhi, Systems Engineering
Product	CSI-100 Cell Sorting Instrument, all configurations

Purpose

End-to-end risk management for the CSI-100 per ISO 14971:2019: plan, hazard identification, risk estimation and evaluation, risk controls and their verification, residual risk, benefit-risk analysis, risk management report, and production and post-production activities. Maintained for the product life.

Document Control

Rev	Date	Phase	Description of change	Author
A	Dec 2024	Concept	Risk management plan; preliminary hazard analysis HA-01 to HA-12.	A. Lodhi
B	Jan 2026	Design	Added HA-13 to HA-15; HA-03 merged into HA-04; DFMEA; software risk analysis; risk controls allocated to SRS-CSI-001 Rev C.	A. Lodhi
C	May 2027	Pre-market release	Added HA-16 (cybersecurity) and HA-17 (specimen misidentification); PFMEA; verification of risk controls; V&V anomalies evaluated; overall residual risk, benefit-risk analysis and risk management report (RMR-CSI-001 Rev A).	A. Lodhi

Approvals

Function	Role	Signature / Date
Quality Engineering	Risk management owner	
Systems Engineering	Author	
Clinical / Medical Affairs	Clinical risk and benefit assessment	
R&D Engineering	Design risk controls	
Software Engineering	Software risk	
Human Factors	Use-related risk	
Manufacturing Engineering	Process risk	
Top management	Approval of risk acceptability policy and overall residual risk	

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1. Introduction and File Structure

This Risk Management File documents risk management for the CSI-100 Cell Sorting Instrument according to ISO 14971:2019 and ISO/TR 24971:2020. It covers the complete system: sorter, fluidics supply module, workstation software and firmware, options and accessories, through design, production and post-production. It is a living file: it is updated whenever design changes, production data or post-production information affect a risk estimate or a risk control.

ISO 14971 clause	Content	Location
4.4 Risk management plan	Scope, responsibilities, acceptability criteria, activities	Section 2 (full plan RMP-CSI-001 Rev B)
5.2, 5.3 Intended use and safety characteristics	Intended use, misuse, characteristics	Section 3
5.4, 5.5 Hazards, hazardous situations, risk estimation	Hazard analysis HA-CSI-001 Rev C	Sections 4 and 5
6 Risk evaluation	Acceptability of each risk	Section 5
7.1 to 7.6 Risk control	Options, implementation, verification, residual risk, new risks	Sections 5, 6, 7, 8, 9
8 Overall residual risk	Evaluation and benefit-risk	Sections 12, 13
9 Risk management review	Risk management report RMR-CSI-001 Rev A	Section 15
10 Production and post-production	Information collection, review and actions	Section 16

2. Risk Management Plan Summary

Element	Plan
Scope	CSI-100 system, all configurations, full product life cycle
Responsibilities	Quality Engineering owns the file; Systems Engineering leads analysis; R&D, Software, Human Factors, Manufacturing, Clinical Affairs and Service contribute; top management approves the acceptability policy
Competence	Cross-functional team with flow cytometry, laser safety, high voltage, biosafety, software and usability expertise
Acceptability criteria	Risk matrix in Section 4.3, derived from the manufacturer's risk acceptability policy, state of the art for cell sorters and applicable standards
Verification of controls	Implementation and effectiveness verified under VVP-CSI-001 Section 8
Overall residual risk	Evaluated by a panel including an independent clinical expert (Section 12)
Production and post-production	Complaints, service data, internal instrument logs, field actions, literature and adverse event databases (Section 16)
Reviews	At each design review, before design transfer, before release and at least annually after release

3. Intended Use and Safety Characteristics

3.1 Intended use

The CSI-100 is an automated, multi-laser flow cytometer and droplet cell sorter intended for the identification, enumeration and physical separation of cell populations in human specimens stained with fluorochrome-labeled reagents, for in vitro diagnostic use by trained laboratory professionals in clinical laboratories, with IVD reagents and assays validated for the instrument.

3.2 Reasonably foreseeable misuse

- Use with reagents or assays not validated for the instrument, or reporting research-mode results as diagnostic.
- Running IVD assays after a failed or skipped QC check, or overriding the QC lockout.
- Opening doors during sorting or before aerosol evacuation after a clog.
- Sorting infectious specimens without the aerosol option or a biosafety enclosure.
- Using detergents or disinfectants not listed in the IFU; filling containers with the wrong fluid.
- Service by untrained persons, including removal of laser shielding.

3.3 Safety characteristics (ISO/TR 24971 Annex A)

Question	Answer for the CSI-100	Related hazards
What energy is delivered or present?	Class 3B lasers, up to 12 kV deflection voltage, mains voltage, compressed air up to 100 PSI, moving parts	HA-01, 02, 05, 06, 09, 12
Is it used with biological materials?	Human specimens, potentially infectious; aerosols generated during sorting	HA-04, HA-10
Does it produce a measurement or result used for diagnosis?	Yes: population identification and counts; sorted fractions for further analysis	HA-07, HA-08, HA-13, HA-17
Does it depend on software?	Yes: Class B software system (SRS-SW-CSI-010)	Section 7
Is it connected to networks?	Workstation to instrument link; optional LIS and network storage	HA-16
Are consumables or fluids used?	Sheath, bleach, ethanol, water; beads and reagents	HA-10, HA-14, HA-15
Is it installed, moved or maintained?	Installed and serviced by trained engineers; 185 kg	HA-11
Could use errors cause harm?	Yes: see critical tasks in Section 8	Section 8

4. Risk Analysis Methods and Criteria

4.1 Methods

Method	Scope	Record
Preliminary hazard analysis	System hazards from intended use and safety characteristics	HA-CSI-001 (Section 5)
Design FMEA	Subsystem failure modes	DFMEA-CSI-001 (summary Section 10)
Process FMEA	Manufacturing process failure modes	PFMEA-CSI-001 (summary Section 10)
Use FMEA and task analysis	Use errors in critical tasks	Usability engineering file (summary Section 8)
Software failure analysis	Contribution of software items to hazardous situations	Section 7; SAD-SW-CSI-011 §10
Threat model	Cybersecurity threats and their effect on safety	Section 9

4.2 Severity and probability

Severity	Definition (patient, operator or other persons)
S5 Catastrophic	Death
S4 Critical	Permanent impairment or life-threatening injury; incorrect result likely to cause misdiagnosis of a serious disease
S3 Serious	Injury requiring medical intervention; incorrect result likely to cause delayed or inappropriate treatment with reversible harm; invasive specimen re-collection
S2 Minor	Temporary injury not requiring medical intervention; repeat of a non-invasive specimen collection
S1 Negligible	Inconvenience; repeat test on the same specimen

Probability of harm ($P = P1 \times P2$)	Per instrument-year of use
P5 Frequent	≥ 1 in 10
P4 Probable	1 in 100 to 1 in 10
P3 Occasional	1 in 1,000 to 1 in 100
P2 Remote	1 in 10,000 to 1 in 1,000
P1 Improbable	< 1 in 10,000

Probability combines the probability that the hazardous situation occurs (P1) and that it leads to harm (P2). For software, the probability of a software failure is taken as 1, per IEC 62304 and IEC/TR 80002-1; the risk estimate relies on P2 and on risk controls outside the failing software item.

4.3 Risk acceptability matrix

Probability \ Severity	S1	S2	S3	S4	S5
P5 Frequent	A	U	U	U	U
P4 Probable	A	R	U	U	U
P3 Occasional	A	A	R	U	U
P2 Remote	A	A	A	R	R
P1 Improbable	A	A	A	A	R

U = unacceptable (dark); R = acceptable only if further reduction is not practicable and benefit outweighs risk (light); A = acceptable (white). Under ISO 14971:2019 and EN ISO 14971:2019+A11:2021, all risks are reduced as far as possible without adversely affecting the benefit-risk ratio, including those in the A region.

5. Hazard Analysis

5.1 Hazard register

ID	Hazard	Harm	Initial S / P	Initial	Residual S / P	Residual
HA-01	Laser radiation	Retinal injury or skin burn.	S4 / P4	Unacceptable	S4 / P1	Acceptable
HA-02	High voltage on deflection plates	Electric shock or burn.	S4 / P4	Unacceptable	S4 / P1	Acceptable
HA-03	Retired: sample splash during loading	—	—	—	—	Retired
HA-04	Infectious aerosol from droplet sorting	Laboratory-acquired infection.	S4 / P3	Unacceptable	S4 / P2	Review (benefit-risk)
HA-05	Access to sort chamber during sorting	Electric shock or exposure to aerosol.	S4 / P3	Unacceptable	S4 / P1	Acceptable
HA-06	Moving parts	Pinch or crush injury.	S2 / P4	Review (benefit-risk)	S2 / P2	Acceptable
HA-07	Wrong cells in sorted fraction	Incorrect downstream result leading to delayed or inappropriate treatment.	S3 / P4	Unacceptable	S3 / P2	Acceptable
HA-08	Incorrect result from configuration, QC or calibration	Misdiagnosis or delayed diagnosis of a serious disease.	S4 / P4	Unacceptable	S4 / P2	Review (benefit-risk)
HA-09	Overpressure or sudden pressure release	Impact injury or fluid spray.	S3 / P3	Review (benefit-risk)	S3 / P1	Acceptable
HA-10	Decontamination agents and residual contamination	Chemical irritation or infection.	S3 / P3	Review (benefit-risk)	S3 / P1	Acceptable
HA-11	Instrument instability	Crush injury.	S3 / P2	Acceptable	S3 / P1	Acceptable
HA-12	Mains electrical shock or fire	Electric shock, burns or death.	S5 / P3	Unacceptable	S5 / P1	Review (benefit-risk)
HA-13	Electromagnetic interference	Incorrect result leading to delayed or inappropriate treatment.	S3 / P3	Review (benefit-risk)	S3 / P1	Acceptable
HA-14	Fluid misconnection	Loss of specimen requiring re-collection, or incorrect result.	S3 / P3	Review (benefit-risk)	S3 / P1	Acceptable
HA-15	Fluid run-out during a sort	Specimen re-collection (for example repeat bone marrow aspiration) and delayed diagnosis.	S3 / P4	Unacceptable	S3 / P2	Acceptable
HA-16	Unauthorized change or loss of data (cybersecurity)	Misdiagnosis or delayed diagnosis.	S4 / P3	Unacceptable	S4 / P1	Acceptable
HA-17	Specimen or result misidentification	Misdiagnosis of two patients.	S4 / P3	Unacceptable	S4 / P2	Review (benefit-risk)

5.2 Risk profile before and after controls

Before risk control (number of hazards in each cell):

Probability \ Severity	S1	S2	S3	S4	S5
P5 Frequent	A	U	U	U	U
P4 Probable	A	R (1)	U (2)	U (3)	U
P3 Occasional	A	A	R (4)	U (4)	U (1)
P2 Remote	A	A	A (1)	R	R
P1 Improbable	A	A	A	A	R

After risk control (residual risk):

Probability \ Severity	S1	S2	S3	S4	S5
P5 Frequent	A	U	U	U	U
P4 Probable	A	R	U	U	U
P3 Occasional	A	A	R	U	U
P2 Remote	A	A (1)	A (2)	R (3)	R
P1 Improbable	A	A	A (5)	A (4)	R (1)

16 active hazards. Before control, 10 were unacceptable. After control, 0 are unacceptable and 4 (HA-04, HA-08, HA-12, HA-17) are in the review region and are justified by the benefit-risk analysis in Section 13.

5.3 Hazard records

Control types follow the priority order of ISO 14971 clause 7.1: ID = inherent safety by design, PM = protective measure in the device or manufacturing process, IS = information for safety.

HA-01	Laser radiation (Energy: optical)
Sequence of events	Operator or service person opens the flow cell access door or a service panel while lasers are on; direct or reflected Class 3B beam becomes accessible.
Hazardous situation	Eye or skin exposed to laser radiation (375 to 633 nm, up to 50 mW).
Harm	Retinal injury or skin burn.
Initial risk	S4 Critical; P4 Probable; Unacceptable
Risk controls	ID: Fully enclosed optical path; Class 1 laser product in all user-accessible conditions (SYS-511) PM: Hardware shutter blocks all beams when the flow cell door opens; firmware rejects conflicting commands (SYS-512, FW-502) IS: Laser warning labels on every panel with Class 3B radiation behind it; IFU restricts service to trained engineers (SYS-513, SYS-1103, SYS-1102)
Software items involved	SI-10
Verification	SVR-CSI-001 (IEC 60825-1 accessible emission); DVP-CSI-502; ANM-0032 label correction verified
Residual risk	S4 Critical; P1 Improbable; Acceptable
New risks from controls	None

HA-02	High voltage on deflection plates (Energy: electrical)
Sequence of events	Operator reaches into the sort block while plate voltage is on (plates stay energized with the sort block door open), or insulation fails, or software leaves voltage on after a fault.
Hazardous situation	Contact with plates at up to 12,000 V.
Harm	Electric shock or burn.
Initial risk	S4 Critical; P4 Probable; Unacceptable
Risk controls	ID: Insulation and guarding so no energized part is accessible with the sort block door closed (SYS-523) PM: Hardware interlock and hardwired e-stop remove HV; independent safety supervisor removes HV enable on missed heartbeat or watchdog (SYS-521, SYS-532, FW-604, FW-605) PM: Firmware safe state on readback fault or link loss; latched until acknowledged (FW-307, FW-603, FW-606, FW-607) PM: Red warning light and on-screen indicator whenever plate voltage is on (SYS-522, RA-1003) IS: HV warning labels; IFU instruction not to touch plates while the warning light is on (SYS-1103)
Software items involved	SI-10, SI-12, SI-15, SI-01; software requirements derived from this hazard: FW-103, FW-307, FW-601, FW-603, FW-604, FW-605, FW-606, FW-607, RA-1003, RA-403
Verification	DVP-CSI-503 (59 trials per response); fault injection in ITS-CSI-010 §10; ANM-0007 closed
Residual risk	S4 Critical; P1 Improbable; Acceptable
New risks from controls	Safety supervisor could remove HV during normal operation (nuisance trip): evaluated, no hazard (ANM-0007)

HA-04	Infectious aerosol from droplet sorting (Biological)
Sequence of events	A nozzle clog or deflected stream generates aerosol in the sort chamber; operator opens the sort block door before aerosol is cleared, or aerosol escapes during sorting.
Hazardous situation	Operator inhales aerosol from a potentially infectious specimen.
Harm	Laboratory-acquired infection.
Initial risk	S4 Critical; P3 Occasional; Unacceptable
Risk controls	ID: Closed sort block and collection chamber doors; sealed collection holders (SYS-521, SYS-804) PM: Continuous breakoff monitoring and clog detection within 500 ms; autonomous firmware stops stream and HV within 2 s (SYS-541, SYS-543, FW-210, FW-602) PM: Aerosol evacuation option with ULPA filter; warning if not running (SYS-551, SYS-552, RA-208) PM: Guided clog recovery: boost evacuation before door opening; restart only after acknowledgement (RA-504, RA-505) IS: IFU biosafety guidance, clog handling and decontamination; recommendation to use the aerosol option or a biosafety enclosure for infectious specimens (SYS-554, SYS-1102)
Software items involved	SI-13, SI-10, SI-01, SI-03; software requirements derived from this hazard: FW-210, FW-602, FW-607, RA-1002, RA-504, RA-505
Verification	DVP-CSI-503, DVP-CSI-504 aerosol challenge; VAL-CSI-03 (critical task); ANM-0011 and ANM-0051 closed
Residual risk	S4 Critical; P2 Remote; Review (benefit-risk)
New risks from controls	None

HA-05	Access to sort chamber during sorting (Energy / biological)
Sequence of events	Operator opens the collection chamber door during sorting to change tubes.
Hazardous situation	Hand near energized plates and aerosol.
Harm	Electric shock or exposure to aerosol.
Initial risk	S4 Critical; P3 Occasional; Unacceptable
Risk controls	ID: Collection chamber door cannot open unless the sort block door is closed (SYS-521) PM: Firmware monitors door state and rejects stream and HV commands that conflict (FW-502) IS: IFU tube change procedure (SYS-1102)
Software items involved	SI-10
Verification	DVP-CSI-503 door sequence test
Residual risk	S4 Critical; P1 Improbable; Acceptable
New risks from controls	None
HA-06	Moving parts (Energy: mechanical)
Sequence of events	Hand or clothing near the loading port, plate stage or aspirator drawer while it moves.
Hazardous situation	Finger trapped by moving part.
Harm	Pinch or crush injury.
Initial risk	S2 Minor; P4 Probable; Review (benefit-risk)
Risk controls	ID: Pinch force below IEC 61010-1 limits (SYS-562) PM: Illuminated e-stop stops all motion; button lights while the port moves; stall detection (SYS-531, SYS-532, SYS-533, FW-501, FW-601, FW-309) IS: Pinch-point labels and IFU warning (SYS-1103)
Software items involved	SI-10, SI-12; software requirements derived from this hazard: FW-601, RA-1002
Verification	DVP-CSI-503, DVP-CSI-505 pinch force and e-stop timing
Residual risk	S2 Minor; P2 Remote; Acceptable
New risks from controls	None
HA-07	Wrong cells in sorted fraction (Incorrect output)
Sequence of events	Breakoff drifts, drop delay is wrong or sort timing is late; drops with non-target cells are sorted, or rare-event sorting is run at too high a rate.
Hazardous situation	Sorted fraction has lower purity than reported and is used for further diagnostic analysis.
Harm	Incorrect downstream result leading to delayed or inappropriate treatment.
Initial risk	S3 Serious; P4 Probable; Unacceptable
Risk controls	ID: Deterministic sort decision in FPGA within 20 μ s (FW-204, SYS-461) PM: Breakoff monitoring pauses charging within 100 ms when out of range (SYS-542, FW-205) PM: Automatic drop delay determination; purity and single-cell precision modes (SYS-465, RA-407, SYS-464) PM: Sort report records settings, pauses and conflicts (SYS-635, RA-706) IS: IFU guidance on event rate for rare populations and on re-analysis of sorted fractions (SYS-1102)
Software items involved	SI-11, SI-13, SI-03; software requirements derived from this hazard: FW-205
Verification	DVP-CSI-405 purity and yield; ITS-CSI-010 §11; VAL-CSI-02; ANM-0003 and ANM-0056 closed
Residual risk	S3 Serious; P2 Remote; Acceptable
New risks from controls	None

HA-08	Incorrect result from configuration, QC or calibration (Incorrect output)
Sequence of events	Instrument runs with a configuration that does not match installed optics, with failed or expired QC, with overdue calibration, or with altered assay settings.
Hazardous situation	Clinical result generated with wrong channel assignment, drifted sensitivity or unapproved analysis.
Harm	Misdiagnosis or delayed diagnosis of a serious disease.
Initial risk	S4 Critical; P4 Probable; Unacceptable
Risk controls	ID: Configuration-driven optics generated from a controlled record with rules; board inventory check at startup (SYS-623, AD-105, AD-106, FW-302) PM: QC lockout: IVD templates blocked without a passing performance check in the last 24 h (SYS-622, RA-803) PM: Configuration match check before acquisition; calibration due check blocks IVD use (RA-805, AD-102, AD-903) PM: Locked, approved IVD assay templates; results flagged when warnings were active (SYS-704, RA-901, RA-903) PM: Keyed and marked tube holders, nozzles and collection devices (SYS-903) IS: IFU requires assay controls with each run and result review by qualified staff (SYS-1102)
Software items involved	SI-05, SI-20, SI-21, SI-10, SI-02; software requirements derived from this hazard: AD-102, AD-104, AD-203, AD-602, AD-903, FW-102, FW-302, FW-304, FW-606, FW-803, FW-807, RA-402, RA-603, RA-701, RA-803, RA-805, RA-901, RA-903
Verification	DVP-CSI-601, DVP-CSI-701; STS-CSI-010 §A and §C seeded-error tests; VAL-CSI-01, VAL-CSI-04; ANM-0016, 0025, 0026 closed
Residual risk	S4 Critical; P2 Remote; Review (benefit-risk)
New risks from controls	QC lockout could delay urgent testing: supervisor override limited to research use (RA-804)

HA-09	Overpressure or sudden pressure release (Energy: pneumatic)
Sequence of events	Regulator fault over-pressurizes tanks, or a tank is opened while pressurized.
Hazardous situation	Tank or line ruptures or cap releases under pressure.
Harm	Impact injury or fluid spray.
Initial risk	S3 Serious; P3 Occasional; Review (benefit-risk)
Risk controls	ID: Tanks rated at 1.5 times maximum pressure with mechanical relief valves (SYS-561) PM: Pressure regulation; system vents within 10 s of power off (FW-401, FW-403, SYS-524) IS: IFU refill and autoclave instructions (SYS-1102)
Software items involved	SI-14; software requirements derived from this hazard: RA-204
Verification	DVP-CSI-505 relief valve and depressurization tests
Residual risk	S3 Serious; P1 Improbable; Acceptable
New risks from controls	None

HA-10	Decontamination agents and residual contamination (Chemical / biological)
Sequence of events	Materials degrade from repeated bleach and ethanol exposure, or decontamination is incomplete before service.
Hazardous situation	Leak of contaminated fluid, or service person touches contaminated parts.
Harm	Chemical irritation or infection.
Initial risk	S3 Serious; P3 Occasional; Review (benefit-risk)
Risk controls	ID: Wetted and exposed materials compatible with 10% bleach and 70% ethanol (SYS-553) PM: Automated cleaning modes (SYS-603, RA-203) IS: IFU decontamination procedure before service (SYS-554)
Software items involved	SI-04
Verification	DVP-CSI-504 material compatibility after 500 cycles
Residual risk	S3 Serious; P1 Improbable; Acceptable
New risks from controls	None

HA-11	Instrument instability (Energy: mechanical)
Sequence of events	Instrument pushed or bench bumped during installation or use.
Hazardous situation	Instrument (up to 185 kg) tips.
Harm	Crush injury.
Initial risk	S3 Serious; P2 Remote; Acceptable
Risk controls	ID: Stable when tilted 10° (SYS-563) IS: Installation only by service engineers; site preparation guide (SYS-1203, SYS-1104)
Software items involved	—
Verification	DVP-CSI-505 tilt test
Residual risk	S3 Serious; P1 Improbable; Acceptable
New risks from controls	None

HA-12	Mains electrical shock or fire (Energy: electrical)
Sequence of events	Insulation failure, protective earth failure or component overheating.
Hazardous situation	Accessible conductive part becomes live, or fire starts.
Harm	Electric shock, burns or death.
Initial risk	S5 Catastrophic; P3 Occasional; Unacceptable
Risk controls	ID: IEC 61010-1 and -2-101 design; fluidics module powered only from sorter (SYS-501, SYS-314) PM: Circuit breaker; rail and temperature monitoring put the instrument in the safe state (SYS-315, FW-308, FW-606) PM: Production earth continuity, dielectric strength and touch current test on every unit (SYS-502)
Software items involved	SI-12, SI-10; software requirements derived from this hazard: FW-308, FW-606
Verification	SVR-CSI-001 (IEC 61010); FTS-CSI-001 §5.16 production test
Residual risk	S5 Catastrophic; P1 Improbable; Review (benefit-risk)
New risks from controls	None

HA-13	Electromagnetic interference (Energy: electromagnetic)
Sequence of events	External fields couple into detector signals.
Hazardous situation	Noise events alter counts and percentages.
Harm	Incorrect result leading to delayed or inappropriate treatment.
Initial risk	S3 Serious; P3 Occasional; Review (benefit-risk)
Risk controls	ID: Immunity per IEC 61326-2-6; shielded detector cabling (SYS-503) PM: Daily QC detects abnormal event rates (SYS-621)
Software items involved	SI-11
Verification	SVR-CSI-001 (IEC 61326) immunity with performance monitoring; ANM-0021 closed
Residual risk	S3 Serious; P1 Improbable; Acceptable
New risks from controls	None
HA-14	Fluid misconnection (Chemical / incorrect output)
Sequence of events	Cleaning fluid connected to the sheath line, or containers swapped.
Hazardous situation	Cells exposed to bleach or ethanol during acquisition.
Harm	Loss of specimen requiring re-collection, or incorrect result.
Initial risk	S3 Serious; P3 Occasional; Review (benefit-risk)
Risk controls	ID: Keyed, non-interchangeable connectors (SYS-801, SYS-905) IS: Color-coded, labeled containers; IFU (SYS-905, SYS-1102)
Software items involved	—
Verification	DVP-CSI-801, DVP-CSI-901 misconnection attempts
Residual risk	S3 Serious; P1 Improbable; Acceptable
New risks from controls	None
HA-15	Fluid run-out during a sort (Loss of function)
Sequence of events	Sheath runs out or waste fills during a long sort.
Hazardous situation	Sort stops; irreplaceable specimen partly consumed.
Harm	Specimen re-collection (for example repeat bone marrow aspiration) and delayed diagnosis.
Initial risk	S3 Serious; P4 Probable; Unacceptable
Risk controls	PM: Level monitoring; warnings at 20% sheath and 80% waste; sort blocked if capacity is insufficient (SYS-493, RA-207, FW-402) ID: 10 L sheath and waste capacity (SYS-492)
Software items involved	SI-04, SI-14; software requirements derived from this hazard: RA-1002
Verification	DVP-CSI-408 low-level scenarios
Residual risk	S3 Serious; P2 Remote; Acceptable
New risks from controls	None

HA-16	Unauthorized change or loss of data (cybersecurity) (Information security)
Sequence of events	Network attack, malicious update, replayed command, or unauthorized user alters settings, templates or results.
Hazardous situation	Clinical results or instrument settings altered without detection.
Harm	Misdiagnosis or delayed diagnosis.
Initial risk	S4 Critical; P3 Occasional; Unacceptable
Risk controls	ID: Locked workstation image, isolated instrument link, signed updates (SYS-707, CM-201, CM-202, FW-104) PM: Authenticated pairing, session keys, CRC and sequence checks (FW-802, FW-803) PM: User authentication, roles, audit trail, electronic signatures, encrypted data at rest (SYS-701, SYS-702, SYS-703, SYS-709, DM-202, DM-504) PM: SBOM and vulnerability management (SYS-708, CM-203)
Software items involved	SI-40, SI-41, SI-10, SI-31, SI-32
Verification	DVP-CSI-701 including penetration test; ANM-0030 closed
Residual risk	S4 Critical; P1 Improbable; Acceptable
New risks from controls	None

HA-17	Specimen or result misidentification (Use error)
Sequence of events	Operator enters the wrong specimen ID, or a result is attached to the wrong record during transcription.
Hazardous situation	Result reported for the wrong patient.
Harm	Misdiagnosis of two patients.
Initial risk	S4 Critical; P3 Occasional; Unacceptable
Risk controls	PM: IVD results require review and release before reporting; release by electronic signature (DM-301, SYS-703) PM: Direct export of released results to the LIS avoids manual transcription (DM-404) PM: Results show template, version and QC status for review (RA-902) IS: IFU and training on specimen identification and review (SYS-1102)
Software items involved	SI-30, SI-05
Verification	VAL-CSI-03 critical task; VAL-CSI-09; STS-CSI-010 §D
Residual risk	S4 Critical; P2 Remote; Review (benefit-risk)
New risks from controls	None

6. Risk Control Verification

Each risk control is verified for implementation (the control is present as specified) and effectiveness (the control reduces the risk as intended). Verification is planned in VVP-CSI-001 Section 8 and traced in TM-CSI-DT-01 Matrix D. All verification results are Pass.

Hazard	Controls (requirements)	Implementation and effectiveness evidence	Result
HA-01 Laser radiation	SYS-511, SYS-512, SYS-513, SYS-1102, SYS-1103, FW-502	SVR-CSI-001 (IEC 60825-1 accessible emission); DVP-CSI-502; ANM-0032 label correction verified	Pass (after ANM-0032)
HA-02 High voltage on deflection plates	SYS-521, SYS-522, SYS-523, SYS-532, SYS-1103, RA-1003, FW-307, FW-603, FW-604, FW-605, FW-606, FW-607	DVP-CSI-503 (59 trials per response); fault injection in ITS-CSI-010 §10; ANM-0007 closed	Pass (after ANM-0007, ANM-0023, ANM-0048)
HA-04 Infectious aerosol from droplet sorting	SYS-521, SYS-541, SYS-543, SYS-551, SYS-552, SYS-554, SYS-804, SYS-1102, RA-208, RA-504, RA-505, FW-210, FW-602	DVP-CSI-503, DVP-CSI-504 aerosol challenge; VAL-CSI-03 (critical task); ANM-0011 and ANM-0051 closed	Pass (after ANM-0011, ANM-0051)
HA-05 Access to sort chamber during sorting	SYS-521, SYS-1102, FW-502	DVP-CSI-503 door sequence test	Pass
HA-06 Moving parts	SYS-531, SYS-532, SYS-533, SYS-562, SYS-1103, FW-309, FW-501, FW-601	DVP-CSI-503, DVP-CSI-505 pinch force and e-stop timing	Pass
HA-07 Wrong cells in sorted fraction	SYS-461, SYS-464, SYS-465, SYS-542, SYS-635, SYS-1102, RA-407, RA-706, FW-204, FW-205	DVP-CSI-405 purity and yield; ITS-CSI-010 §11; VAL-CSI-02; ANM-0003 and ANM-0056 closed	Pass (after ANM-0003, ANM-0056)
HA-08 Incorrect result from configuration, QC or calibration	SYS-622, SYS-623, SYS-704, SYS-903, SYS-1102, RA-803, RA-805, RA-901, RA-903, FW-302, AD-102, AD-105, AD-106, AD-903	DVP-CSI-601, DVP-CSI-701; STS-CSI-010 §A and §C seeded-error tests; VAL-CSI-01, VAL-CSI-04; ANM-0016, 0025, 0026 closed	Pass (after ANM-0012, ANM-0016, ANM-0025, ANM-0026, ANM-0030, ANM-0040, ANM-0060)
HA-09 Overpressure or sudden pressure release	SYS-524, SYS-561, SYS-1102, FW-401, FW-403	DVP-CSI-505 relief valve and depressurization tests	Pass
HA-10 Decontamination agents and residual contamination	SYS-553, SYS-554, SYS-603, RA-203	DVP-CSI-504 material compatibility after 500 cycles	Pass
HA-11 Instrument instability	SYS-563, SYS-1104, SYS-1203	DVP-CSI-505 tilt test	Pass
HA-12 Mains electrical shock or fire	SYS-314, SYS-315, SYS-501, SYS-502, FW-308, FW-606	SVR-CSI-001 (IEC 61010); FTS-CSI-001 §5.16 production test	Pass
HA-13 Electromagnetic interference	SYS-503, SYS-621	SVR-CSI-001 (IEC 61326) immunity with performance monitoring; ANM-0021 closed	Pass (after ANM-0021)
HA-14 Fluid misconnection	SYS-801, SYS-905, SYS-1102	DVP-CSI-801, DVP-CSI-901 misconnection attempts	Pass
HA-15 Fluid run-out during a sort	SYS-492, SYS-493, RA-207, FW-402	DVP-CSI-408 low-level scenarios	Pass

Hazard	Controls (requirements)	Implementation and effectiveness evidence	Result
HA-16 Unauthorized change or loss of data (cybersecurity)	SYS-701, SYS-702, SYS-703, SYS-707, SYS-708, SYS-709, FW-104, FW-802, FW-803, DM-202, DM-504, CM-201, CM-202, CM-203	DVP-CSI-701 including penetration test; ANM-0030 closed	Pass
HA-17 Specimen or result misidentification	SYS-703, SYS-1102, RA-902, DM-301, DM-404	VAL-CSI-03 critical task; VAL-CSI-09; STS-CSI-010 §D	Pass

7. Software Risk Management

Software risk management follows IEC 62304 clause 7. The software system is Class B; the rationale depends on hardware controls independent of software (laser shutter interlock, hardwired HV removal, relief valves, containment hardware) and on external controls for incorrect results (assay controls, QC, result review).

Hazard	Software items that can contribute	Software risk controls	Segregation / independence
HA-02	SI-10, SI-12	Safe state, readback checks, watchdog (FW-307, FW-603 to FW-607)	SI-15 on separate MCU removes HV enable; hardware interlocks
HA-04	SI-13, SI-10, SI-01	Clog detection and autonomous response (FW-210, FW-602); guided recovery (RA-504)	Response in firmware, independent of workstation
HA-07	SI-11, SI-13, SI-03	Deterministic FPGA sort path; breakoff-OK gating (FW-204, FW-205)	Discrete breakoff-OK signal, independent of software messages
HA-08	SI-05, SI-20, SI-21, SI-10, SI-02	QC lockout, configuration rules and inventory, calibration due, assay lock	External: assay controls, laboratory review
HA-15	SI-04, SI-14	Level monitoring and capacity check	SI-14 Class A segregated on fluidics MCU
HA-16	SI-40, SI-41, SI-10, SI-31, SI-32	Authentication, signed updates, session keys, audit trail	Isolated instrument link
HA-17	SI-30, SI-05	Review and release, LIS export	External: laboratory identification procedures

Software requirements derived from hazards: 41 links from 8 hazards to SRS-SW-CSI-010. SOUP anomalies are evaluated in the SOUP register; the only SOUP anomaly found in V&V (ANM-0044, plotting library memory growth) had no safety effect.

8. Use-Related Risk

Use-related risk is managed under IEC 62366-1. Critical tasks were identified from the use FMEA and evaluated in the summative usability study VAL-CSI-03 (15 intended users per user group, final labeling, production-equivalent units).

Critical task	Potential use error	Hazard	Summative result
Loading a sample tube	Hands in loading port while moving	HA-06	Pass
Clog recovery	Opening sort block door before aerosol evacuation	HA-04	Pass after ANM-0051 (guided procedure)
Changing collection tubes during sort	Opening collection door while HV on	HA-05	Pass
Nozzle change and setup selection	Setup for wrong nozzle size	HA-08	Pass after ANM-0060 (nozzle marking)
Daily QC and review of QC status	Running IVD assay with failed QC	HA-08	Pass
Fluid refill and connection	Connecting wrong fluid container	HA-14	Pass
Decontamination after infectious specimens	Incomplete decontamination	HA-10	Pass
Result review and release	Releasing result for wrong specimen	HA-17	Pass

9. Cybersecurity Risk

Security risks are assessed with a threat model (IEC 81001-5-1, AAMI TIR57). Threats that can affect safety or effectiveness are carried into HA-16; controls are verified by DVP-CSI-701, including penetration testing.

Threat	Effect on safety	Controls	Residual
Malicious software update	Altered sort or analysis behavior	Signed updates and manifest (FW-104, CM-202)	Acceptable
Spoofed or replayed commands	HV or stream commands from an attacker	Pairing, session keys, sequence and CRC (FW-802, FW-803); ANM-0030 closed	Acceptable
Unauthorized user	Altered templates or results	Authentication, roles, audit trail (SYS-701, SYS-702)	Acceptable
Network malware on workstation	Loss or corruption of data	Locked image, minimal services, isolated link (SYS-707, CM-201)	Acceptable
Tampering with exported data	Wrong result in LIS	Checksums on export (DM-402)	Acceptable

10. FMEA Summaries

10.1 Design FMEA (highest-risk failure modes)

Subsystem	Failure mode	Effect	S	P before	Controls	P after	Hazard
SS-04 Excitation optics	Beam spot shifts after shipping	Reduced sensitivity; wrong result	4	3	Transport-validated crate; alignment check at installation; QC	1	HA-08
SS-03 Flow cell	Optical gel void	High CVs; wrong result	3	3	Gel process validation; CV test in FAT; QC	1	HA-08
SS-07 Sort mechanism	Charge timing error	Low purity	3	3	FPGA deterministic timing; purity test	1	HA-07
SS-07 Sort mechanism	HV insulation breakdown	Shock	4	2	Insulation design; 100% hipot; interlock	1	HA-02
SS-09 Aerosol management	Chamber seal leak	Aerosol escape	4	3	Seal design; containment test each unit	1	HA-04
SS-01 Fluidics module	Regulator fails open	Overpressure	3	3	Relief valve; pressure monitoring	1	HA-09
SS-06 Electronics	PMT HV drift	Sensitivity drift	3	3	HV readback fault; QC	1	HA-08
SS-02 Sample delivery	Loading port motor stall	Pinch or stuck tube	2	3	Stall detection; e-stop	2	HA-06

10.2 Process FMEA (highest-risk failure modes)

Process controls are validated and transferred under DTP-CSI-001; ECO references show changes made during V&V.

Process step	Failure mode	S	P before	Controls	P after	Hazard
WI-CSI-1002 Gel coupling	Void in gel	3	3	Validated dispense and cure; void inspection (ECO-0187)	1	HA-08

Process step	Failure mode	S	P before	Controls	P after	Hazard
WI-CSI-1003 Launch plate alignment	Spot out of tolerance	4	3	Validated alignment; FTS-CSI-001 §3	1	HA-08
WI-CSI-1005 Detector arrays	Wrong filter for configuration	4	3	Kitting scan; configuration rules; FAT spectral check	1	HA-08
WI-CSI-1007 HV assembly	Insulation damage	4	2	Validated assembly; 100% hipot	1	HA-02
Aerosol chamber sealing	Seal mis-seated	4	3	Validated assembly; containment test each unit	1	HA-04
WI-CSI-1009 Software install	Wrong version or configuration file	4	2	Checksum and configuration readback	1	HA-08
Labeling	Missing warning label	4	2	Label map per build; inspection (ECO-0193)	1	HA-01
WI-CSI-1012 Packing	Inadequate crating	3	3	Validated crate (ECO-0197)	1	HA-08

11. Risks from Anomalies Found in V&V

Every anomaly in ADL-CSI-001 related to a hazard was reviewed by Quality for its effect on the risk estimate and on risk controls.

Anomaly	Hazard	Effect on risk	Outcome
ANM-0003 Sort decision latency exceeds 20 μ s above 95,000 events/s	HA-07	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0007 Safety supervisor removes HV enable during normal boot	HA-02	No change to risk estimate	Corrected and re-verified; risk estimate unchanged after correction
ANM-0011 Clog response exceeds 2 s in 2 of 59 trials	HA-04	Risk control failed to meet requirement	Corrected and re-verified; risk estimate unchanged after correction; HA-CSI-001 updated
ANM-0012 PI CV 3.1% on DV-02 on day 2	HA-08	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0016 Laser calibration factor applied to wrong laser on shared beam spot	HA-08	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0021 Radiated immunity: event rate spikes at 3 V/m, 80 to 120 MHz	HA-13	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0023 After link loss, Research App shows Ready while firmware is in safe state	HA-02	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0025 QC lockout window 25 h across daylight saving change	HA-08	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0026 480/20 filter rule not enforced after field upgrade to 445 nm	HA-08	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0030 Replayed command accepted after reconnect (penetration test)	HA-08	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0032 Laser warning label missing on rear panel of 5-laser build	HA-01	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0040 DV-05 needed realignment after ISTA 3E transport test	HA-08	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0048 Duplicate of ANM-0023	HA-02	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0051 Usability verification: clog recovery step skipped	HA-04	No change to risk estimate	Corrected and re-verified; risk estimate unchanged after correction; HA-CSI-001 updated
ANM-0056 VAL-CSI-02 Site 3: CD34+ purity 93% on two specimens	HA-07	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0060 VAL-CSI-03: users unsure which nozzle is installed	HA-08	No change to risk estimate	Corrected and re-verified; risk estimate unchanged after correction; HA-CSI-001 updated
ANM-0066 Plate voltage readback drift warning after 6 h sorting	HA-02	No change to risk estimate	Corrected and re-verified; evaluated; open - fix verified

12. Overall Residual Risk Evaluation

The overall residual risk was evaluated by a panel of Quality, Systems, Clinical Affairs, Human Factors and an independent clinical flow cytometry expert, using the method in RMP-CSI-001:

- Review of every individual residual risk (Section 5) and of the risk controls' verification (Section 6).

- Consideration of combined risks: hazards affecting the same user in the same task (HA-02, HA-04, HA-05 during sorting) and hazards affecting the same result (HA-07, HA-08, HA-13, HA-17).
- Comparison with legally marketed cell sorters and flow cytometers, their labeling and published adverse event data.
- Review of anomalies from V&V (Section 11) and validation findings.
- Review of the residual risks disclosed in labeling (Section 14).

Conclusion: no individual residual risk is unacceptable. 4 residual risks are in the review region and are justified in Section 13. The overall residual risk is judged acceptable in relation to the benefits of the intended use and is comparable to or lower than that of similar devices on the market.

13. Benefit-Risk Analysis

13.1 Benefits

- Accurate identification and enumeration of cell populations for diagnosis and monitoring of hematological malignancies and immune disorders (VAL-CSI-01, VAL-CSI-04).
- Physical isolation of rare or specific populations at high purity and viability for further diagnostic analysis, such as molecular testing of sorted cells, which improves the sensitivity of downstream tests (VAL-CSI-02).
- Single-cell deposition with index data, enabling clonality and single-cell analyses in the clinical laboratory.
- Automated setup, QC and result control reduce operator-dependent variability compared with manual workflows.

13.2 Residual risks in the review region

Hazard	Residual risk	Justification
HA-04	Aerosol exposure remains possible if infectious specimens are sorted without the aerosol option or a biosafety enclosure.	Droplet sorting inherently generates aerosol; further reduction without the option would require enclosure of every instrument, which would limit use in many laboratories. Controls exceed those of comparable sorters; option and biosafety guidance disclosed.
HA-08	An incorrect result could still occur if a failure is not caught by QC or assay controls.	Multiple independent controls (configuration rules, QC lockout, calibration check, assay lock, assay controls, laboratory review). Remaining probability is remote; benefit of accurate automated cytometry outweighs risk.
HA-12	Death from electric shock or fire is possible in principle.	Design meets IEC 61010-1 and -2-101 with 100% production testing; probability improbable; comparable to all mains-powered laboratory instruments.
HA-17	Specimen misidentification by the user cannot be fully prevented by the instrument.	Instrument supports review, release and LIS export without transcription; specimen identification is controlled by laboratory procedures. Barcode reading is planned for a future release.

13.3 Conclusion

The medical benefits of the CSI-100 when used as intended outweigh the overall residual risk. Each residual risk in the review region has been reduced as far as possible without adversely affecting the benefit-risk ratio, and is disclosed to users.

14. Disclosure of Residual Risks

Residual risk	Where disclosed
Laser radiation behind service panels	Labels (LBL-CSI-001); OG-CSI-001 safety chapter; SM-CSI-001
High voltage on deflection plates with sort block door open	HV label and warning light; OG-CSI-001
Aerosol exposure when sorting infectious specimens	OG-CSI-001 biosafety chapter; recommendation for aerosol option or biosafety enclosure
Incorrect results if assay controls and QC are not used	OG-CSI-001 intended use, limitations and QC chapter
Reduced purity for rare populations at high event rates	OG-CSI-001 sorting chapter (event rate guidance)
Specimen misidentification	OG-CSI-001 result review chapter
Moving parts	Pinch-point labels; OG-CSI-001
Cybersecurity	OG-CSI-001 and customer security documentation (SBOM available on request)

15. Risk Management Report

This section is the risk management report RMR-CSI-001 Rev A required by ISO 14971 clause 9, reviewed before commercial release of the CSI-100.

Review question (clause 9)	Finding	Evidence
Has the risk management plan been implemented appropriately?	Yes	Sections 2 to 11; RMP-CSI-001 Rev B
Is the overall residual risk acceptable?	Yes	Sections 12 and 13
Are appropriate methods in place to collect and review production and post-production information?	Yes	Section 16; complaint handling and service procedures
Have all risk controls been implemented and verified?	Yes	Section 6; VVP-CSI-001; TM-CSI-DT-01 Matrix D
Have new risks from risk controls been evaluated?	Yes	Section 5.3, 'New risks from controls'
Have V&V anomalies related to hazards been evaluated?	Yes	Section 11; ADL-CSI-001
Are open items acceptable for release?	Yes: ANM-0066 (HA-02) evaluated, no change in risk; closure review 2027-05-20	ADL-CSI-001 Section 9

Conclusion of the review: the CSI-100 risk management process meets ISO 14971:2019; the overall residual risk is acceptable; the device may be released for commercial distribution, subject to the closure review noted above.

Reviewer	Role	Signature / Date
Quality Engineering	Risk management owner	
Clinical Affairs	Clinical assessment	
Systems Engineering	Technical lead	
Independent reviewer	Clinical flow cytometry expert	
Top management	Release approval	

16. Production and Post-Production Activities

This file is maintained throughout the life of the CSI-100. The following information is collected, reviewed for new hazards or changed risk estimates, and fed back into this file (ISO 14971 clause 10).

Source	Content	Review
Complaints and adverse event reports	Customer complaints; reportable events (MDR, vigilance)	Each event; monthly trend
Service data	Repairs, replaced parts, field alignment results	Monthly
Instrument internal logs	Safe-state entries, clog events, HV faults, usage counters (FW-701 to FW-704)	Quarterly from service log exports
Production data	NCRs, first-pass yield, FAT failures, process capability	Monthly (DTP-CSI-001 Section 15)
Cybersecurity monitoring	Vulnerabilities in SOUP and the SBOM	Monthly
External sources	Literature, adverse event databases, recalls of similar devices, standards updates	Annually

Triggers for updating the file: any new hazard or hazardous situation; any harm occurring more often or more severely than estimated; any failed or bypassed risk control; any design or process change; changes in the state of the art. Updates are reviewed through the same process and recorded in the document control table.

17. Reference Documents

Document	Title
ISO 14971:2019; ISO/TR 24971:2020	Application of risk management to medical devices; guidance
IEC 62304; IEC/TR 80002-1	Software life cycle; guidance on applying ISO 14971 to software
IEC 62366-1	Usability engineering
IEC 81001-5-1; AAMI TIR57	Security activities in the product life cycle; security risk management
IEC 61010-1, -2-101; IEC 60825-1; IEC 61326-2-6	Electrical, laser and EMC safety
RMP-CSI-001 Rev B	Risk Management Plan
SRS-CSI-001 Rev D; SRS-SW-CSI-010 Rev E	System and Software Requirements Specifications
SAD-SW-CSI-011 Rev C	Software Architecture Description
VVP-CSI-001 Rev A; TM-CSI-DT-01 Rev C	V&V Plan; Requirements Traceability Matrix
ADL-CSI-001 Export 07	Anomaly and Defect Tracking Log
DFMEA-CSI-001 Rev C; PFMEA-CSI-001 Rev A	Design and Process FMEA
DTP-CSI-001 Rev B	Design Transfer Plan
OG-CSI-001 Rev C; LBL-CSI-001 Rev B	Operator's Guide; Labeling Specification